

Smart Home AIoT – Test Data Collection Sheet – Phase A – Single Device Verification

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Instructions for use:

1. Run each test 10 times consecutively (or across sessions, as practical)
2. Record: Date, Time, exact Input given, Actual Output observed, Response Time (use stopwatch), and Pass/Fail
3. Pass criteria: Actual matches Expected description
4. After 10 trials, calculate: Pass Rate %, Average Response Time, Standard Deviation
5. Use Comments for any anomalies or special observations

Statistical formulas:

- Pass Rate = $(\text{Pass Count} / 10) \times 100\%$
- Avg RT = $(\text{response times}) / 10$
- Variance = $(x_i -)^2 / n$
- Std Dev = $\sqrt{\text{variance}}$

Coverage Matrix

Phase	Description	Count
Phase A	Phase A – Single Device Verification	16
	TOTAL	16

Phase A – Single Device Verification

A01	Bedroom Light	Phase A					
Method: HA toggle switch.bedroom_light							
Expected: ไฟห้องนอนติด/ดับ + state เปลี่ยน							
#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10		Pass Rate: ____%		Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

Method: HA toggle light.living_room_light

Expected: Tuya bulb ON with color default

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

A03

Kitchen Light

Phase A

Method: HA toggle switch.kitchen_light

Expected: ไฟครัวติด/ดับ

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: HA toggle switch.bathroom_light

Expected: ไฟห้องน้ำติด/ดับ

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms
Comments:			

A05

Garden Light

Phase A

Method: HA toggle switch.garden_light

Expected: ไฟสวนหลักติด/ดับ

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

A06

Garage Light

Phase A

Method: HA toggle switch.garage_light

Expected: ไฟจราจรหลักติด/ดับ

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

A07

Garage Pathway Light

Phase A

Method: HA toggle switch.garage_pathway_light

Expected: ไฟทางเดินโรงรถ

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: HA toggle switch.garden_pathway_light

Expected: ไฟทางเดินสวน

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

A09

Bedroom Fan

Phase A

Method: HA toggle switch.bedroom_fan

Expected: พัดลมห้องนอนหมุน

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: HA toggle switch.living_room_fan

Expected: พัดลมห้องนั่งเล่นหมุน

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

Method: HA toggle switch.kitchen_fan

Expected: vent fan ครัวทำงาน

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

A12

Bedroom Curtain – Open

Phase A

Method: HA cover.bedroom_curtain open

Expected: ม่านพับเก็บขึ้น (หน้าต่างโล่ง)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: HA cover.bedroom_curtain close

Expected: ม่านลงคลุมหน้าต่าง

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms
Comments:			

Method: HA lock.front_door_lock

Expected: servo หมุน 090°

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms
Comments:			

A15

Garage Door

Phase A

Method: HA cover.garage_door open/close

Expected: servo หมุนเปิด/ปิด

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

A16

Garden Water Pump

Phase A

Method: HA toggle switch.garden_water_pump

Expected: ปั๊มทำงาน + น้ำไหล

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments: